

Estados estacionarios

$$V(x, t) ; \textcircled{V(x)}$$

Son soluciones de la ecuación de Schrödinger en donde el potencial no depende del tiempo.

$$i\hbar \frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} + V(x) \psi(x, t)$$

Ecuación de Schrödinger

Busquemos soluciones en variables separables:

$$\psi(x, t) = \varphi(x) \phi(t)$$

$$\frac{1}{\varphi \phi} \left(i\hbar \frac{\partial \varphi \phi}{\partial t} = -\frac{\hbar^2}{2m} \frac{\partial^2 \varphi \phi}{\partial x^2} + V(x) \varphi \phi \right)$$
$$E \quad i\hbar \frac{\partial \phi}{\phi \partial t} + \underbrace{\left(\frac{\hbar^2}{2m} \frac{1}{\varphi} \frac{\partial^2 \varphi}{\partial x^2} - \frac{V(x) \varphi}{\varphi} \right)}_{\text{depende de } x} = 0 \quad -E$$

depende de t

$$i\hbar \frac{\partial \phi}{\partial t} = E \phi \quad (1)$$

$$-\frac{\hbar^2}{2m} \frac{\partial^2 \varphi}{\partial x^2} + V(x) \varphi = E \varphi \quad (2)$$

$$\overbrace{\left(-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + V(x) \right)}^H \varphi = E \varphi$$

Ecuación estacionaria de Sch.

$$\text{De (1): } \phi(t) = A e^{-\frac{i}{\hbar} E t}$$

La dependencia temporal se reduce a una fase

$$\rho(x, t) = |\psi(x, t)|^2 = |\varphi(x)|^2 |\phi(t)|^2$$
$$= |A|^2 |\varphi(x)|^2 \quad (\text{independiente del tiempo})$$

$$\psi(x, t) = \varphi(x) e^{-\frac{i}{\hbar} E t}$$

Una partícula de masa m está en el estado

$$\psi(x,t) = A e^{-a[(\frac{m}{\hbar}x^2) + it]}$$

A y a son ctes > 0

a) Encontrar A

b) ¿Qué potencial $V(x)$ satisface la ecuación Sch.?

c) x, x^2, p, p^2

d) Δx y Δp

$$\int_{-\infty}^{\infty} |\psi|^2 dx = 1 \rightarrow |A|^2 \int_{-\infty}^{\infty} e^{-\frac{2amx^2}{\hbar}} dx = |A|^2 \sqrt{\frac{\pi\hbar}{2ma}}$$

$$A = \left(\frac{2ma}{\pi\hbar}\right)^{1/4}$$

$$\psi(x) = \left(\frac{2ma}{\pi\hbar}\right)^{1/4} e^{-amx^2/\hbar}$$

Debe satisfacer la ecuación de Sch.

$$-\frac{\hbar^2}{2m} \frac{d^2\psi(x)}{dx^2} + V(x)\psi(x) = E\psi(x)$$

$$i\hbar \frac{\partial \psi(x,t)}{\partial t} = E\psi(x,t) \rightarrow E = \hbar a$$

$$-\frac{\hbar^2}{2m} \frac{d^2\psi(x)}{dx^2} = \left(\frac{2ma}{\pi\hbar}\right)^{1/4} a(\hbar - 2amx^2) e^{-amx^2/\hbar}$$

$$V(x) = \frac{E\psi(x) + \frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2}}{\psi(x)} = 2ma^2 x^2$$

oscilador

$$\langle x \rangle_{\psi} = 0$$

$$\langle x^2 \rangle_{\psi} = \int_{-\infty}^{\infty} x^2 \psi^2 dx = \hbar/4am$$

$$\langle p \rangle_{\psi} = -i\hbar \int_{-\infty}^{\infty} \psi \frac{\partial \psi}{\partial x} dx = 0, \langle p^2 \rangle_{\psi} = -\hbar^2 \int_{-\infty}^{\infty} \psi \frac{\partial^2 \psi}{\partial x^2} dx = \hbar ma$$

$$\int_{-\infty}^{\infty} e^{-ax^2} = \sqrt{\frac{\pi}{a}}$$

Prob 1.9 Griffiths pag. 32

$$i\hbar \frac{\partial \psi}{\partial t} = i\hbar(-ai)\psi = E\psi \rightarrow E = \hbar a$$

PIH

$$i\hbar \frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} + V(x)\psi(x,t)$$

$$\Delta x = \frac{1}{2} \sqrt{\frac{\hbar}{ma}}$$

$$\Delta p = \sqrt{\hbar ma}$$

$$\Delta x \Delta p = \frac{\hbar}{2}$$

$$\Delta x = \sqrt{\langle x^2 \rangle_{\psi} - \langle x \rangle_{\psi}^2}$$

$$\Delta p =$$

Supongamos una partícula en $t=0$ se encuentra en una combinación de dos estados estacionarios

$$\psi(x,0) = c_1 \psi_1(x) + c_2 \psi_2(x)$$

Encuentre la función de onda $\psi(x,t)$

$$\psi(x,t) = c_1 \psi_1(x) e^{-i \frac{E_1}{\hbar} t} + c_2 \psi_2(x) e^{-i \frac{E_2}{\hbar} t}$$

donde E_1 y E_2 son las energías asociadas con ψ_1 y ψ_2

$$\begin{aligned} |\psi(x,t)|^2 &= \left(c_1^* \psi_1^* e^{i \frac{E_1}{\hbar} t} + c_2^* \psi_2^* e^{i \frac{E_2}{\hbar} t} \right) \left(c_1 \psi_1 e^{-i \frac{E_1}{\hbar} t} + c_2 \psi_2 e^{-i \frac{E_2}{\hbar} t} \right) \\ &= |c_1|^2 |\psi_1|^2 + c_1^* \psi_1^* c_2 \psi_2 e^{\frac{i}{\hbar} (E_1 - E_2) t} + c_2^* c_1 \psi_2^* \psi_1 e^{-\frac{i}{\hbar} (E_1 - E_2) t} + |c_2|^2 |\psi_2|^2 \end{aligned}$$

En el caso $c_1, c_2, \psi_1, \psi_2 \in \mathbb{R}$

$$= c_1^2 \psi_1^2 + c_2^2 \psi_2^2 + 2 c_1 c_2 \psi_1 \psi_2 \cos\left(\frac{E_1 - E_2}{\hbar} t\right)$$

No es un estado estacionario

Hamiltoniano Cuántico

Se define como

$$H \equiv -\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + V(x)$$

Con esto, la ec. estacionaria de Sch. sería una ec. de vectores y valores propios.

$$H\varphi = E\varphi$$

$$\varphi \in L^2(-\infty, \infty)$$

$H\varphi_n = E_n\varphi_n$ ← Sol. espacial de la ec. estacionaria

$E_n \longrightarrow \varphi_n$ $\{E_n\}$ espectro de energía

↑ ↑ ↑
(Energía)(Estado propio)($\in \mathbb{R}$)

$\{\varphi_n(x)\}$ Base del espacio ...

Se demuestra que corresponde a un sistema ortonormal.

$H = H^+$ auto adjunto/hermítico

$$(\varphi_n, \varphi_m) = \int_{-\infty}^{\infty} dx \varphi_n^* \varphi_m = \delta_{nm}$$

$$\delta_{nm} = \begin{cases} 1 & \text{si } n = m \\ 0 & \text{si } n \neq m \end{cases}$$

$$\psi(x, t) = \sum_n \alpha_n \underbrace{\varphi_n(x)}_{\text{Sol. Estacionario de Sch.}} e^{-\frac{i}{\hbar} E_n t}$$

↑
(Función de estado) ($\in \mathbb{C}$)

(Paquete de onda)

(No tiene una energía propia)

$$\psi^*(x, t) = \sum_n \alpha_n^* \varphi_n^*(x) e^{\frac{i}{\hbar} E_n t}$$

$$|\psi(x, t)|^2 = \sum_{n, m} \alpha_n^* \alpha_m \underbrace{\varphi_n^*(x) \varphi_m(x)}_{\delta_{nm}} e^{\frac{i}{\hbar} (E_n - E_m) t}$$

$$\|\psi\|^2 = \int_{-\infty}^{\infty} |\psi(x, t)|^2 dx = \sum_{n, m} \alpha_n^* \alpha_m \int_{-\infty}^{\infty} \underbrace{\varphi_n^*(x) \varphi_m(x)}_{\delta_{nm}} dx e^{\frac{i}{\hbar} (E_n - E_m) t} = \sum_n |\alpha_n|^2 = 1$$

BORN

$$H = \frac{p^2}{2m} + V$$

$$= \sum_n |\alpha_n|^2 = 1 \quad (m = n)$$

$$\psi(x, t) \xrightarrow{\text{Ec.Sch}} \psi(x, t) \begin{array}{l} \nearrow \varphi_1 \\ \rightarrow \varphi_2 \\ \searrow \vdots \\ \varphi_n \end{array}$$

$$H\varphi_i = E_i \varphi_i$$

$$\varphi_1 \rightarrow E_1 \rightarrow |\alpha_1|^2$$

$$\varphi_2 \rightarrow E_2 \rightarrow |\alpha_2|^2$$

$\vdots$

$$\varphi_n \rightarrow E_n \rightarrow |\alpha_n|^2$$

$$\langle E \rangle_\psi = \sum_{i=1}^n |\alpha_i|^2 E_i$$

$$\psi(x, t) = \sum_n \alpha_n \varphi_n(x) e^{-\frac{i}{\hbar} E_n t}$$

$$\psi(x, 0) = \sum_n \alpha_n \varphi_n(x)$$

$$(\varphi_m, \psi(x, 0)) = \sum_n \alpha_n \underbrace{(\varphi_m, \varphi_n)}_{\delta_{mn}}$$

$$= \alpha_m$$

Como el subíndice es mudo:

$$\alpha_n = (\varphi_n, \psi(x, 0)) \in \mathbb{C}$$

$$|\alpha_n|^2: \text{probabilidad}$$

Valores esperados:

X: observable posición

- $X = X^+$
- Sus valores propios son los resultados de una medición.
- Sus estados propios son los estados a los que SALTA la función de estado, cuando se efectúa la medición.

$$\langle \hat{X} \rangle_\psi = \int_{-\infty}^{\infty} dx \, x \rho(x) : \text{valor medio de la posición}$$

↑
(Variable posición)

$$\rho(x) = |\psi|^2$$

$$\begin{aligned} \langle \hat{X} \rangle_\psi &= \int_{-\infty}^{\infty} dx \, x \psi^* \psi \\ \text{(Depende de la función de onda)} &= \int_{-\infty}^{\infty} dx \, \psi^* x \psi = (\psi, \hat{X} \psi) \end{aligned}$$

$$\hat{X} \psi = x \psi$$

- El operador posición es un operador multiplicativo.
 $x\psi = x\psi$

$$\begin{aligned} & f(\hat{x}) \quad \text{Ejm: } f(\hat{x}) = \hat{x}^2 + \hat{x} \\ & \downarrow \quad \quad \quad \uparrow \text{(observable)} \\ \langle f(\hat{x}) \rangle_\psi &= \int_{-\infty}^{\infty} dx \, \psi^*(x,t) f(x) \psi(x,t) \\ & \quad \quad \quad \uparrow \text{variable} \\ & = (\psi, f(\hat{x}) \psi) \end{aligned}$$

$$\begin{aligned} f(\hat{x}) &= \hat{x}^2 + \hat{x} \\ f(\hat{x})\psi &= \hat{x}^2 \psi + \hat{x} \psi \\ &= x^2 \psi + x \psi \\ &= f(x) \psi \end{aligned}$$

Tengamos cuidado con la identificación del operador hamiltoniano:

$$H\psi + \frac{\hbar}{i} \partial_t \psi = 0$$

$$H\psi = -\frac{\hbar}{i} \partial_t \psi, \quad \psi(x,t) \text{ (no es solución arbitraria)}$$

$$i\hbar \frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} + V(x) \psi(x,t)$$

$$H \equiv -\frac{\hbar}{i} \partial_t \psi \quad \hat{H} \psi = -\frac{\hbar}{i} \partial_t \psi \quad \hat{H} \rightarrow -\frac{\hbar}{i} \partial_t$$

$$\hat{H} \psi = \left(-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + V(x) \right) \psi$$

$$\hat{H} \psi \rightarrow i\hbar \frac{\partial}{\partial t} \psi$$

Partícula libre

$$V(t, x) = 0$$

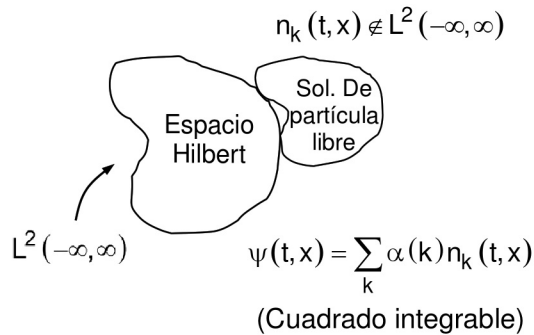
$$\psi = n(t, x) = e^{i(kx - \frac{E}{\hbar}t)}$$

(muestre que es solución de la ec. de Sch.)

Se verifica si

$$E = \frac{(\hbar k)^2}{2m}$$

$$|n_k(t, x)|^2 = 1$$



Observable Momentum:

$$-i\hbar \frac{\partial}{\partial x} e^{\frac{i}{\hbar}px} = -i\hbar \frac{i}{\hbar} p e^{\frac{i}{\hbar}px} = p e^{\frac{i}{\hbar}px}$$

$\left\{ e^{\frac{i}{\hbar}px} \right\}$: estados propios del operador momentum.

$$\boxed{P \equiv -i\hbar \frac{\partial}{\partial x}} \quad P = P^+ \rightarrow \text{Valores propios reales}$$

Solo si la función de onda es una onda plana, el sistema tendrá momentum definido (propio).

$$\underbrace{\psi(x, t)}_{\in L^2(-\infty, \infty)} = \sum_k \alpha_k n_k(t, x)$$

$$\boxed{i\hbar \frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} + V(x) \psi(x, t)}$$

$$i\hbar \frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2}$$

$$i\hbar \frac{\partial \psi}{\partial t} = i\hbar \left(-\frac{iE}{\hbar} \right) \psi$$

$$-\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} = -\frac{\hbar^2}{2m} i^2 k^2 \psi$$

$$E = \frac{\hbar^2 k^2}{2m}$$

$\hat{p}$

(Ec. Sch)

$$\psi(x,0) \rightarrow \psi(x,t) \begin{cases} \eta_1 \rightarrow P_1 \rightarrow |\alpha_1|^2 \\ \eta_2 \rightarrow P_2 \rightarrow |\alpha_2|^2 \\ \vdots \\ \eta_k \rightarrow P_k \rightarrow |\alpha_k|^2 \end{cases}$$

$$\langle P \rangle_\psi = \sum_i |\alpha_i|^2 P_i$$

$$i\hbar \frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} + V(x) \psi(x,t)$$

$$\underbrace{\psi(x,t)}_{\in L^2(-\infty, \infty)} = \sum_k \alpha_k \eta_k(t, x)$$

Primera Cuantificación:

$$f_{\text{operador}} = f\left(x, -i\hbar \frac{\partial}{\partial x}, t\right)$$

Observable físico:

- Operador hermítico:

$$(\Omega f, g) = (f, \Omega^+ g) \quad , \quad \Omega = \Omega^+$$

- Operador lineal:

$$\Omega(\alpha f + \beta g) = \alpha \Omega f + \beta \Omega g$$

$$\langle f_{\text{operador}} \rangle_\psi = \int_{-\infty}^{\infty} \psi^*(x, t) f_{\text{operador}} \psi(x, t) dx$$

Clásicamente: $h = \frac{p^2}{2m} + v$

Cuánticamente:

$p \longrightarrow \hat{P} = -i\hbar \frac{\partial}{\partial x}$
 valor momentum Operator momentum

$v \longrightarrow V$
Operator potencial

$$h = \frac{p^2}{2m} + v(x, t)$$

Observable físico



Operador
Hermítico

$$\hat{h} \psi = \frac{1}{2m} \left(-i\hbar \frac{\partial}{\partial x} \right) \left(-i\hbar \frac{\partial \psi}{\partial x} \right) + \hat{V} \psi$$

$$= -\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} + \hat{V} \psi$$

$$\hat{h} = -\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + \hat{V}$$

Cuántico

$$\hat{H}_{\text{operador}} = -\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + \hat{V}(x, t)$$

La mecánica cuántica como teoría de las mediciones

Medición:

$$x = a \pm c \quad \text{(incertidumbre)}$$

Mecánica cuántica:

$$\psi(t, \vec{x}) \in \mathbb{C}$$

Función de estado

contiene toda la información

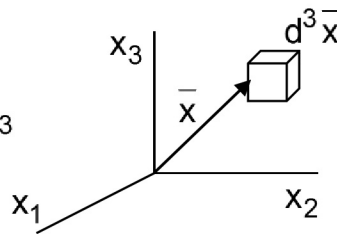
$$\rho(t, \vec{x}) = |\psi(t, \vec{x})|^2 \quad \text{Densidad de probabilidad}$$

$$\in \mathbb{R}^{3+1}$$

$$\iiint_{\text{volumen}} \overbrace{\rho(t, \vec{x}) d^3 \vec{x}}^{\text{probabilidad}} = 1$$

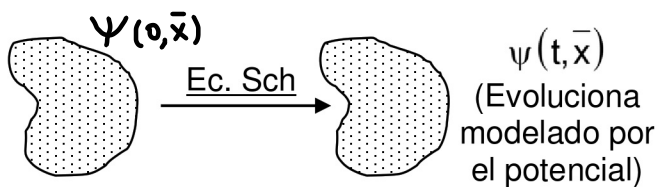
$$\downarrow$$

$$dx_1 dx_2 dx_3$$



$$\begin{aligned} \|\psi\|^2 &= (\psi, \psi) \\ &= \iiint_{-\infty}^{\infty} \underbrace{\psi^* \psi}_{|\psi|^2} dV \end{aligned}$$

$\psi(t_0, \vec{x})$: función de estado inicial



$$-\frac{\hbar^2}{2m} \nabla^2 \psi(t, \bar{x}) + \hat{V}(t, \bar{x}) \psi(t, \bar{x}) + \frac{\hbar}{i} \frac{\partial \psi(t, \bar{x})}{\partial t} = 0$$

Variable de posición

$\frac{\partial}{\partial t} = \frac{\partial}{\partial t}$

$$\nabla^2 = \nabla \cdot \nabla \quad \text{gradiente}$$

divergencia

$$\hbar = \frac{h}{2\pi} \Rightarrow h = 6,63 \times 10^{-34} \text{ J/s}$$

$\nabla \equiv$ nabla

(Ec. diferencial de primer orden en el tiempo)

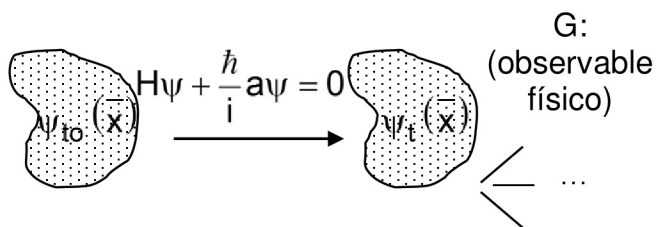
$$\hat{H} = -\frac{\hbar^2}{2m} \nabla^2 + \hat{V} \quad \begin{cases} H = H^+ \\ \text{Lineal} \end{cases}$$

Operador hamiltoniano

$$H\psi + \frac{\hbar}{i} \frac{\partial \psi}{\partial t} = 0$$

(Ec. Sch. dependiente del tiempo)

Proceso de medición:



Estado propio G medida probabilidad

$$\begin{matrix} \varphi_1 & \text{---} & r_1 & \text{---} & |\alpha_1|^2 \\ \varphi_2 & \text{---} & r_2 & \text{---} & |\alpha_2|^2 \\ \vdots & & \vdots & & \vdots \\ \varphi_n & \text{---} & r_n & \text{---} & |\alpha_n|^2 \end{matrix}$$

$$\text{G } \varphi_i = r_i \varphi_i$$

$$\psi_t(\bar{x}) = \sum_{i=1}^n \alpha_i \varphi_i \quad \left(\begin{array}{l} (\in \mathbb{C}) \\ \alpha_i = \alpha_i(t) \\ \varphi_i = \varphi_i(\bar{x}) \end{array} \right)$$

$|\alpha_i|^2$: probabilidad de obtener el valor r_i .

$$\begin{aligned} (\psi_t, \psi_t) &= \left(\sum_i \alpha_i \varphi_i, \sum_j \alpha_j \varphi_j \right) \\ &= \sum_i \sum_j \alpha_i \alpha_j \underbrace{(\varphi_i, \varphi_j)}_{\delta_{ij}} \end{aligned}$$

$$\|\psi_t\|^2 = (\psi_t, \psi_t) = \sum_i |\alpha_i|^2 = 1$$

(Norma²) (También representan Probabilidades)

$$\rho_t = |\psi_t|^2$$

$$\|\psi_t\|^2 = \int |\psi_t|^2 d^3 \bar{x} = 1$$

$$G_{\text{prom}} = \sum_k r_k |\alpha_k(t)|^2 = \sum_i \sum_k r_k \alpha_i^* \alpha_k \delta_{jk}$$

$$= \sum_{j,k} r_k \alpha_j^* \alpha_k (\varphi_j, \varphi_k)$$

$$= \left(\sum_j \alpha_j \varphi_j, \sum_k r_k \alpha_k \varphi_k \right)$$

$$= \left(\sum_j \alpha_j \varphi_j, \sum_k \alpha_k G \varphi_k \right)$$

$$= \left(\sum_j \alpha_j \varphi_j, G \sum_k \alpha_k \varphi_k \right)$$

$$= (\psi, G\psi)$$

$$G \varphi_i = r_i \varphi_i$$

$$\boxed{(\psi_t, G\psi_t) \equiv G_{\text{prom}}, \quad (\psi_t \equiv \psi_t(\bar{x}))}$$

(debe corresponderse con la medida experimental)

Desviación media cuadrática:

$$\begin{aligned} (\delta_\psi G)^2 &= (\psi, (G - G_\psi)^2 \psi) \\ &= (\psi, \{G^2 - GG_\psi - G_\psi G + G_\psi^2\} \psi) \\ &= (\psi, G^2 \psi) - 2G_\psi(\psi, G\psi) + G_\psi^2(\psi, \psi) \\ &= (\psi, G^2 \psi) - G_\psi^2 \\ (\delta_\psi G)^2 &= (G_\psi, G_\psi) - G_\psi^2 \\ &= \|G_\psi\|^2 \end{aligned}$$

$$\|G_\psi\|^2 = (\sigma_\psi G)^2 + G_\psi^2$$

$$\left. \begin{aligned} &\bullet \psi \\ &\bullet (\psi, G(\psi)) = G_{\text{prom}} \\ &\bullet (\sigma_\psi G)^2 = \|G_\psi\|^2 - G_\psi^2 \end{aligned} \right\} G_{\text{prom}} \pm \sigma_\psi G$$

$$G_{\text{prom}} = (\psi, G\psi) = G_\psi$$

ψ : función de estado

$$\begin{aligned} (\sigma_\psi G)^2 &= (\psi, (G - G_\psi)^2 \psi) \\ &= (G^2)_\psi - (G_\psi)^2 \quad \checkmark \end{aligned}$$

Medición: $\boxed{G_\psi \pm \sigma_\psi G}$

$$\begin{aligned} &(\psi, \{G^2 - GG_\psi - G_\psi G + G_\psi^2\} \psi) \\ &= (\psi, G^2 \psi) - (\psi, GG_\psi) \psi - (\psi, G_\psi G \psi) + (\psi, G_\psi^2 \psi) \\ &\quad \underbrace{\hspace{10em}}_{-2G_\psi^2} \quad \underbrace{\hspace{10em}}_{G_\psi^2} \end{aligned}$$

$$(\delta_\psi G)^2 = (\psi, G^2 \psi) - G_\psi^2$$

$$\delta_\psi G = \sqrt{(\psi, G^2 \psi) - G_\psi^2}$$

$$(\delta_\psi G)^2 = \underbrace{(G\psi, G\psi)}_{\|G\psi\|^2} - G_\psi^2$$

$$\|G\psi\|^2 = (\delta_\psi G)^2 + G_\psi^2$$

$$\psi_0 \rightarrow \psi_t$$

$$\begin{aligned} G\psi_t \\ G^2\psi_t \end{aligned}$$

$$G_{\text{prom}} = (\psi, G\psi)$$

Ejm

$\{\varphi_i\}$ base ortonormalizada

La función de estado de un sistema es

$$\psi = (1+i)\varphi_1 + (1-i)\varphi_2 + i\varphi_3$$

$$|\psi|^2 = 5 \quad (\psi, \psi)$$

$$\psi_N = \frac{1+i}{\sqrt{5}}\varphi_1 + \frac{1-i}{\sqrt{5}}\varphi_2 + \frac{i}{\sqrt{5}}\varphi_3$$

Los estados propios y valores propios de un operador G son:

$$G\xi_1 = 2\xi_1$$

$$G\xi_2 = 4\xi_2$$

$$G\xi_3 = 6\xi_3$$

$$\xi_1 = \frac{\varphi_1 - \varphi_2}{\sqrt{2}}$$

$$\xi_2 = \frac{\varphi_1 + \varphi_2}{\sqrt{2}}$$

$$\xi_3 = \varphi_3$$

$$\varphi_1 = \frac{\xi_1 + \xi_2}{\sqrt{2}}$$

$$\varphi_2 = \frac{\xi_2 - \xi_1}{\sqrt{2}}$$

$$\varphi_3 = \xi_3$$

Se pide hallar $(\psi_N, G\psi_N)$

$$(\psi_N, G\psi_N) = -\frac{2}{5}$$

$$\int_{-\infty}^{\infty} \psi_N^* \underbrace{G\psi_N}_{\text{red}} =$$

$$(\psi_N, G\psi_N)$$

$$G\psi_N =$$

$$\frac{1+i}{\sqrt{5}} G\varphi_1 + \frac{1-i}{\sqrt{5}} G\varphi_2 + \frac{i}{\sqrt{5}} G\varphi_3$$